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Studierichting: Informatica
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$$\int_0^1 x\left(x-1\right)^8 dx$$

$$\text{O.I.} = \int x\left(x-1\right)^8 dx \overset{t=1-x}{=} -\int \left(1-t\right) t^8 dt = \int t^9 - t^8 dt$$

$$= \frac{t^{10}}{10} - \frac{t^9}{9} + c \overset{t=1-x}{=} \frac{\left(1-x\right)^{10}}{10} - \frac{\left(1-x\right)^9}{9} + c$$

$$\Rightarrow \text{B.I.} = \left[\frac{\left(1-x\right)^{10}}{10} - \frac{\left(1-x\right)^9}{9}\right]_0^1 = -\left(\frac{1}{10} - \frac{1}{9}\right) = \frac{1}{90}$$

References